

VARDO, Norway

WMO: 010980

Lat: 70.371N

Lon: 31.096E

Elev: 49

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.9	12.6	6.0	7.5	10.8	8.8	8.7	14.3	44.4	24.0	40.9	24.6	21.0	210	0.883

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	62.3	57.0	59.0	54.6	56.3	52.7	57.4	61.6	55.1	58.3	53.2	55.7	9.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.0	64.5	59.6	53.0	60.0	57.1	51.5	56.5	54.9	24.8	61.8	23.3	58.6	22.2	55.8	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.4	32.0	28.4	DB	5.6	68.5	4.5	4.3	2.4	71.6	-0.3	74.1	-2.8	76.5	-6.1	79.6	(4)
(5)			WB	5.3	61.5	4.4	3.1	2.1	63.7	-0.4	65.5	-2.8	67.3	-6.0	69.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	25.3	24.3	27.5	32.4	38.2	44.4	49.9	50.0	46.1	38.1	31.3	28.1	(6)
(7)		DBStd	10.42	6.11	6.43	4.96	5.02	4.89	4.43	4.60	4.12	4.21	5.05	5.03	5.69	(7)
(8)		HDD50	5108	766	721	699	528	366	179	58	51	131	368	562	679	(8)
(9)		HDD65	10452	1231	1141	1164	978	830	618	467	466	568	833	1012	1144	(9)
(10)		CDD50	130	0	0	0	0	1	10	56	50	13	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	13.8	18.1	18.1	16.9	13.2	11.1	10.3	9.4	9.3	11.7	14.1	16.2	17.8	(14)
(15)	Precipitation	PrecAvg	22.8	2.1	1.8	1.6	1.4	1.3	1.7	2.0	2.1	1.9	2.5	2.2	2.2	(15)
(16)		PrecMax	32.7	6.6	4.8	4.9	3.5	3.0	4.3	5.1	5.6	4.5	5.6	6.3	4.9	(16)
(17)		PrecMin	12.4	0.7	0.3	0.2	0.4	0.4	0.2	0.2	0.6	0.6	0.7	0.6	0.2	(17)
(18)		PrecStd	4.0	1.0	0.9	0.8	0.7	0.7	0.9	1.3	1.0	0.7	1.0	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	39.3	40.3	46.5	56.0	62.4	68.5	67.7	59.0	51.6	44.3	41.5	(19)
(20)		0.4%	MCWB	37.2	36.8	36.8	42.1	48.2	56.3	59.9	60.6	53.4	48.7	41.8	38.3	(20)
(21)		2%	DB	37.3	36.8	37.8	42.5	51.0	56.6	63.2	61.6	56.0	49.0	41.0	38.8	(21)
(22)		2%	MCWB	35.5	34.5	35.1	39.1	45.6	52.0	57.8	56.5	52.3	46.3	39.0	36.6	(22)
(23)		5%	DB	35.3	35.0	35.9	40.2	47.8	53.1	59.8	58.2	53.8	46.9	39.3	37.2	(23)
(24)		5%	MCWB	33.3	32.9	33.6	37.6	43.6	49.1	55.6	54.2	50.7	44.7	37.5	35.1	(24)
(25)		10%	DB	33.3	33.0	34.5	38.7	45.2	50.3	56.7	55.8	51.9	45.1	37.8	35.6	(25)
(26)		10%	MCWB	31.4	31.1	32.3	36.5	41.8	46.9	53.4	52.5	49.4	43.1	35.9	33.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	37.2	37.3	42.1	49.1	56.5	61.2	61.2	54.9	49.1	42.2	38.8	(27)
(28)		0.4%	MCDWB	38.9	39.0	39.6	46.1	54.5	61.3	66.8	65.4	57.3	51.0	44.2	40.6	(28)
(29)		2%	WB	35.5	34.9	35.5	39.5	46.1	52.6	58.2	57.1	52.9	46.6	39.3	36.8	(29)
(30)		2%	MCDWB	37.1	36.5	37.5	41.9	50.4	55.9	62.7	60.9	55.3	48.5	40.7	38.5	(30)
(31)		5%	WB	33.6	33.1	34.0	38.0	43.9	49.5	55.9	54.7	51.1	45.0	37.7	35.4	(31)
(32)		5%	MCDWB	34.9	34.8	35.7	39.8	47.4	52.5	59.6	57.5	53.3	46.8	39.1	37.0	(32)
(33)		10%	WB	31.7	31.2	32.5	36.8	42.0	47.3	53.6	53.0	49.7	43.3	36.1	34.0	(33)
(34)		10%	MCDWB	33.1	32.9	34.3	38.5	44.8	49.9	56.3	55.3	51.6	45.0	37.6	35.6	(34)
(35)	Mean Daily Temperature Range	MDBR	7.1	7.1	6.0	5.8	6.1	6.2	6.7	6.5	6.1	5.2	5.9	6.5	(35)	
(36)		5% DB	MCDBR	8.4	7.3	6.6	6.6	11.0	11.7	13.2	11.2	8.7	6.1	5.9	7.0	(36)
(37)		5% DB	MCWBR	7.8	6.7	6.0	5.1	7.4	8.0	8.9	7.8	6.2	5.4	5.7	6.3	(37)
(38)		5% WB	MCDBR	8.2	7.3	6.4	6.0	10.4	11.3	12.6	10.3	7.8	5.8	5.7	6.9	(38)
(39)		5% WB	MCWBR	7.8	6.9	5.9	4.8	7.3	8.1	8.9	7.7	6.0	5.4	5.7	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	N/A	0.265	0.315	0.337	0.352	0.344	0.360	0.346	0.315	0.270	N/A	N/A	(40)	
(41)		taud	N/A	2.087	2.213	2.301	2.383	2.484	2.471	2.491	2.512	2.177	N/A	N/A	(41)	
(42)		Ebn at Noon	0	135	213	251	262	270	259	249	224	131	0	0	(42)	
(43)		Edh at Noon	0	12	23	29	30	28	28	24	18	10	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0	77	424	944	1319	1504	1366	922	485	130	0	0	(44)	
(45)		RadStd	0	10	32	74	145	104	124	91	45	14	0	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+1.21	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page